

MosAlc Lab Assessment

You have been invited to interview for the MosAlc Lab, Mosaic Insurance's AI engineering team! This document is your complete preparation guide. It explains the Lloyd's insurance context you need (no prior knowledge assumed), walks through every data file included in this pack, and gives you a plain-English guide to your case study

DISCLAIMER: THIS IS AN ENTIRELY HYPOTHETICAL CASE STUDY BUILT ON ENTIRELY HYPOTHETICAL NUMBERS AND SCENARIOS

Section 1 — Insurance & Lloyd's Primer

Everything you need to know before you build. Not all this information needs to be used for the assessment, but please use it to help yourself gain context to Mosaic.

1.1 What is specialty insurance?

Insurance is a contract where a policyholder pays a premium and, in return, an insurer agrees to pay if a specified bad event (a 'loss') occurs. Most people know consumer insurance (car, home), however, Mosaic writes specialty insurance high-value, often one-of-a-kind risks that standard insurers won't touch.

Examples of specialty risks Mosaic underwrites:

- A bank's critical systems hit by ransomware: Cyber insurance
- A mining company whose assets are expropriated by a foreign government: Political Risk

1.2 What is Lloyd's of London?

Lloyd's of London is not a single insurance company. It is a marketplace, like a stock exchange, but for insurance. Multiple underwriting syndicates operate within Lloyd's

Lloyd's concept	Plain-English analogy
Lloyd's of London	The marketplace like the London Stock Exchange, but for insurance
Syndicate	A trading entity within Lloyd's like a listed company on the exchange. Mosaic has two Syndicate 1609 and 2610.
Underwriter	The person who decides whether to insure a risk and at what price, like a credit analyst at a bank
Broker	Intermediary who brings risks on behalf of clients, like an estate agent, but for insurance
Submission	A broker sending a risk to an underwriter for consideration, like a loan application
Bound / Placed	The underwriter accepted the risk and the policy is live, like a loan being approved and drawn down
NTU (Not Taken Up)	The underwriter quoted but the broker/client chose a different insurer, like a buyer choosing another estate agent's listing
Declined	Mosaic refused to offer terms, outside appetite or unacceptable risk quality

1.3 Suggested metrics – feel free to add any more as required

Metric	Formula	What it tells you
GWP (Gross Written Premium)	Sum of all premiums written	Top-line revenue: the primary measure of business volume
Hit Rate	$\text{Bound} \div (\text{Bound} + \text{Quoted} + \text{Declined} + \text{NTU})$	How many opportunities convert to policies: a measure of pricing competitiveness and underwriting quality
Loss Ratio	$\text{Claims Incurred} \div \text{GWP}$	What proportion of premium is paid out in claims. Target: below 60% for most lines

Combined Ratio	Loss Ratio + Expense Ratio	Below 100% = underwriting profit. Above 100% = underwriting loss
GWP Plan vs Actual	Actual GWP ÷ Plan GWP	How the business is actually performing against its annual budget/goals. Key driver of management action

1.4 Mosaic's eight Lines Of Business (LoB)

Each line has distinct risk characteristics, typical clients, and data structures. You do not need to be an expert in any of them, but knowing what each one covers will help you make sensible design decisions.

Line	Covers	Typical client & risk
Cyber	Ransomware, data breaches, system failures, regulatory fines	Banks, hospitals, tech companies. Mosaic's fastest-growing line.
Transactional Liability	Losses from false statements made in M&A transactions	Private equity firms, corporates buying or selling businesses
Environmental	Pollution, contamination, remediation liability	Manufacturers, oil companies, real estate developers near legacy industrial sites
Political Risk	Government expropriation, currency restrictions, contract frustration	Multinationals operating in emerging markets; project finance lenders
Political Violence	Physical damage from terrorism, riots, strikes, civil commotion	Hotels, shopping centres, commercial property in risk-prone regions
Financial Institutions	Fraud, crime, D&O liability, cyber for banks specifically	Banks, asset managers, brokers, fintechs
Professional Lines	Negligence and errors in professional services	Law firms, accountants, architects, consultants
Excess Casualty	Large bodily injury and property damage liability above primary limits	Heavy industry, construction, chemicals; policies often \$100m+ limits

Section 2 — The MosAlc Tech Stack

How the AI platform is built

Mosaic has an AI chatbot which is a multimodal AI gateway, which will soon integrate with our data in our AWS data-lake.

2.1 XFi: Policy Data Entry Platform

XFi is Mosaic's system of record for all policies that have been bound (meaning all risks that we have agreed to provide insurance for). In this system we enter all the details of all the risks we have undertaken. Think of it as the CRM and ledger combined: it stores every policy, every endorsement, every premium transaction, and every document associated with a risk from binding to expiry. This is the first point of entry for all our data.

What XFi holds

- Bound policy records (keyed by UMR)
- Premium transactions and bordereaux
- Policy documents (MRC, endorsements)
- Endorsement and amendment history
- Broker and insured contact data

2.2 The Data Lakehouse

The Bronze / Silver / Gold pattern

The home of all the Data entered via XFi. This is our data lakehouse architecture, think of it as three levels of data refinement:

- **Bronze** — Raw data exactly as it arrived. Immutable. A PDF parsed to JSON is Bronze. A premium CSV exported from XFi is Bronze. Never edited.
 - **Silver** — Cleansed, validated, and joined data. Bronze tables from different source systems are joined here, using UMR as the universal key.
 - **Gold** — Business-ready aggregations and metrics. Hit rates, loss ratios, plan vs. actual. This is what executives and underwriters query.
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- **UMR (Unique Market Reference) is the 12-character alphanumeric key that connects every record about a risk across all systems. If you were designing a database for an insurer, UMR is your primary reference key across almost every table. Assume that the key across the Data is the UMR.**

2.3 The Chatbot UI Front End

The front end is the governed multi-model chat interface where Mosaic employees interact with MosAlc agents. Think of it as ChatGPT but controlled by Mosaic, with team workspaces, prompt libraries, and API tool integrations. Your agents will eventually surface in the Chatbot front end

2.4 The AI Gateway

What it is

After the request is entered from the UI front end, it routes through the AI gateway that sits in front of LLM providers (Anthropic, AWS Bedrock, Azure OpenAI). Every request from every Mosaic application routes through this.

Key concepts:

- **Virtual Keys:** You configure them once; the gateway handles rotation and access control.
- **Configs (Routing Rules):** JSON objects that define how requests are routed: primary model, fallback model, load balancing, retry logic.
- **Metadata Tags:** Key-value pairs attached to every request. Used for observability and cost attribution.
- **Prompt Caching:** The AI Gateway + Anthropic can cache large system prompts, reducing tokens billed. If your system prompt is 2,000 tokens and you process 1,000 submissions, this saves ~2m tokens.
- **Semantic Caching:** If two semantically similar requests come in, the Gateway returns the cached response. Useful for repeated analytics queries.

2.5 Semantic Layer

Mosaic's proprietary semantic layer sits above the data Lakehouse and acts as the authoritative translation layer between business questions and structured data queries. When an underwriter asks 'what is our Cyber hit rate this quarter?' The Semantic layer defines exactly what 'hit rate' means, which tables to join, and which users are allowed to see the result.

2.6 AWS Bedrock: Model Hosting

Mosaic runs LLM inference through AWS Bedrock, Amazon's managed model hosting service. Available models include Claude Sonnet (primary), Claude Haiku (automation/speed), Amazon Nova Pro, and Cohere Command R+. For this assessment, use whichever API you have access to, you won't be penalised for using the Anthropic API directly instead of Bedrock.

Section 3 — Data Files Guide

What's in each file and how to use it

Case Study: Underwriting Performance Analyst

File	Description
case4_weekly_submissions.csv	12 weeks of submission activity across all 8 LoB. Columns: week_ending (Sunday), lob, submissions_count, quoted_count, bound_count, declined_count, ntu_count. Calculate hit rate as: $\text{bound_count} / (\text{bound_count} + \text{quoted_count} + \text{declined_count} + \text{ntu_count})$.
case4_weekly_premium.csv	12 weeks of GWP vs. plan data. Columns: week_ending, lob, actual_gwp, plan_gwp, ytd_actual, ytd_plan. The plan figures are pre-loaded; actual figures contain the embedded signals.
case4_pipeline.csv	Open quote pipeline by week and LoB. Columns: week_ending, lob, open_quotes_count, open_quotes_gwp_est, avg_days_in_pipeline. Long avg_days_in_pipeline may indicate broker/pricing friction.
case4_loss_indicators.csv	Weekly attritional loss indicators. Columns: week_ending, lob, new_claims_count, new_claims_incurred_est, attritional_loss_ratio_ytd. Loss ratios are YTD figures — they accumulate across the 12 weeks.

The four signals hidden in the Case data — find all four:

- **Signal 1 — Excess Casualty below plan:** actual GWP is consistently 52–70% of plan across all 12 weeks. This is a structural underperformance, not a one-off blip. The CUO needs to understand why are we losing to price? Are we being too selective?
- **Signal 2 — Cyber hit rate collapse (weeks 9–12):** hit rate drops from ~25–35% to ~8–14% in the final four weeks. This is a sudden deterioration, suggesting a pricing or market conditions change.
- **Signal 3 — Environmental loss ratio trend:** attritional loss ratio starts near target (~55%) but trends up ~3 percentage points per week from week 6 onwards. By week 12, it may be approaching or exceeding 70%. A deteriorating trend is more alarming than a high static number.
- **Signal 4 — Political Violence outperforming:** actual GWP is 112–135% of plan consistently. This is good news, but the CUO also wants to know whether loss ratios are holding up alongside the growth.

Section 4 — Practical Build Tips

4.1 Use AI tools aggressively

We expect you to use Claude, Copilot, or any other AI coding assistant.

4.2 Start with the data, not the code

Before writing a single line, spend 5 minutes reading the data files and understanding what they contain. The cases are designed so that understanding the data well gives you a significant advantage in the build. Look for the intentional quirks, they're there to test your instincts.

4.3 Stack choice is yours

There is no Mosaic-preferred stack for this assessment. However, be prepared to explain why you chose it.

Case Study— The Underwriting Performance Analyst

5.1 Background

Every Monday morning, Mosaic’s Chief Underwriting Officer (CUO) receives a performance pack: GWP vs. plan, hit rate trends, new business pipeline, and early loss indicators. Today this is assembled manually by a data analyst from four different spreadsheets and takes half a day. The output is a static Excel workbook and a two-page Word document narrative.

Your task is to build an AI-powered weekly performance analyst that ingests structured data, performs the analysis, and produces both an interactive dashboard and an executive narrative — automatically. At every step remember to challenge us on the old ways of doing things and show us what’s possible.

5.2 The Data

The synthetic dataset include:

Dataset	Fields Required
Weekly submissions	week_ending, lob, submissions_count, quoted_count, bound_count, declined_count, ntu_count
Weekly premium	week_ending, lob, actual_gwp, plan_gwp, ytd_actual, ytd_plan
Pipeline	week_ending, lob, open_quotes_count, open_quotes_gwp_est, avg_days_in_pipeline
Loss indicators	week_ending, lob, new_claims_count, new_claims_incurred_est, attritional_loss_ratio_ytd

5.3 The Analyst Agent

Build an agent that:

- Ingests the four datasets (CSV or in-memory)
- Detects the four performance signals in the data, plus any additional anomalies
- Generates a brief executive narrative: what happened this week, what is the trend, and what action is recommended. Written for the CUO, not a data scientist.
- Produces a structured JSON summary that could feed a dashboard
- Flags the top three concerns and top one opportunity, each with a one-line rationale

5.4 The Dashboard

Build a simple interactive HTML dashboard (or Streamlit app, or React component) powered by the JSON output.

It should show:

- GWP vs. plan — 12-week trend by LoB
- Hit rate heatmap — LoB vs. week
- Top 3 concerns and top 1 opportunity, surfaced prominently
- The narrative rendered as formatted text

5.5 The Prompt Engineering Challenge

This case is partly a test of prompting skill. The narrative output will be evaluated on:

- Accuracy: does it correctly identify the signals in the data?
- Tone: is it written as a CUO would want to read it: direct, action-oriented, no hedging?

- Length: Is it to the point and not too wordy ?
- Specificity: does it use actual numbers, not vague language?

Bring your prompt(s) to the presentation

5.6 Deliverables

- Agent code
- Sample executive narrative output
- Dashboard
- Prompt file(s)

5.7 Probing Questions

- The CUO calls you and says 'the narrative last week said Cyber was the top concern but the underwriter says it was a one-off. How do you stop this happening?' What would you check, and how could you make the analysis less likely to over-react to a single week?
- How would you keep track of changes to the prompt that writes the narrative, so that if the style shifts you can see what changed and go back to an earlier version?
- If this report needs to be produced at 6am every Monday with nobody watching, what could go wrong and what would you put in place around the agent?